

COOP PROJECT

ScholarEase (AI-Powered Research Paper
Simplification System)

Project Mentor

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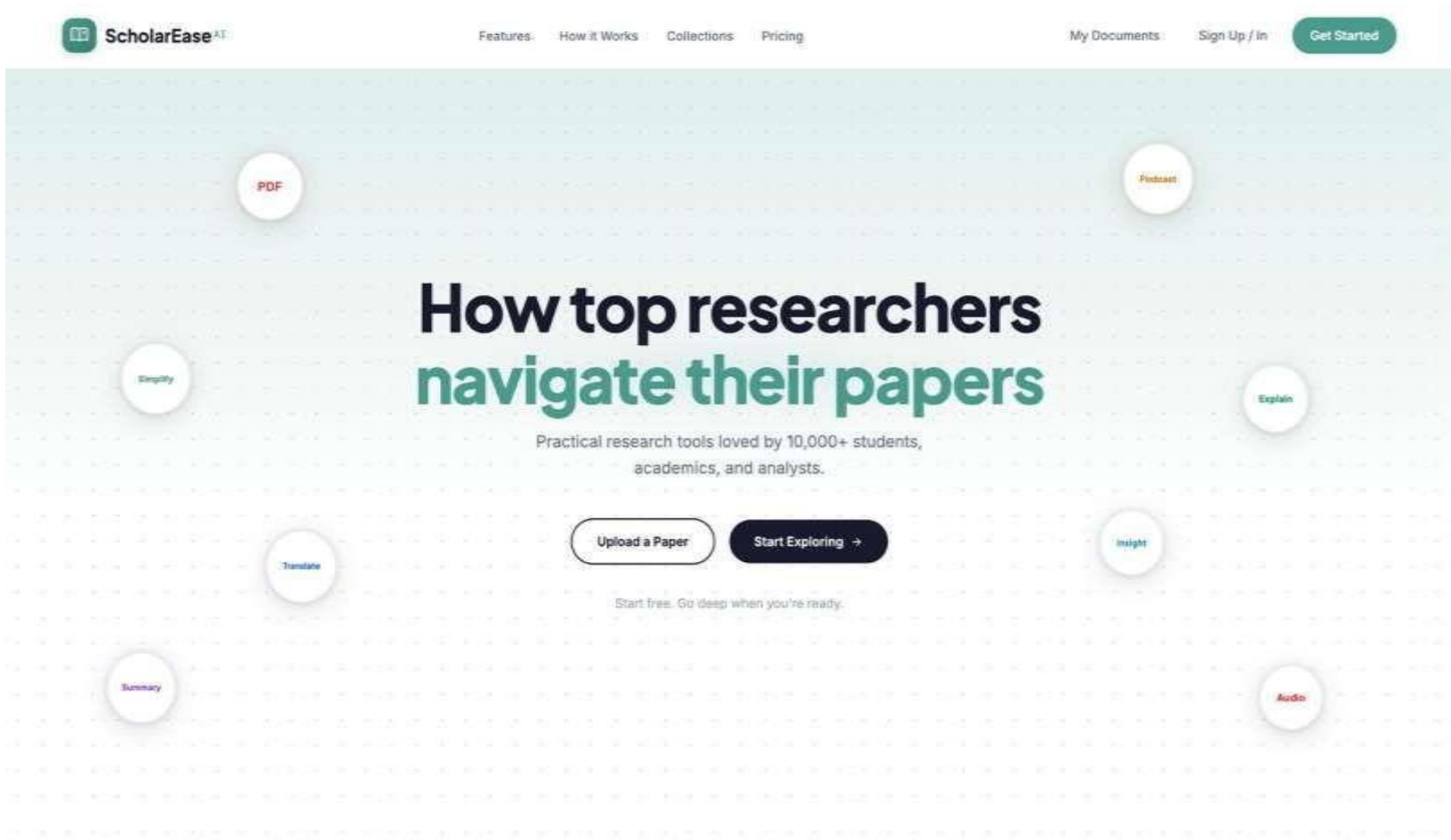
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ScholarEase is an AI-powered web application designed to simplify the analysis of academic research papers. Built using modern technologies such as React.js, Ruby on Rails, and PostgreSQL, it allows users to upload documents and receive concise summaries, keywords, and key insights through an intuitive interface. The platform leverages Natural Language Processing to identify important sections like objectives, methodologies, and conclusions, while also providing visual representations such as concept graphs for better understanding. With features like secure authentication and efficient document management, ScholarEase offers a scalable and user-friendly solution to enhance research productivity and comprehension.

- **React.js** – Frontend library used to build an interactive and responsive user interface.
- **Ruby on Rails** – Backend framework used to develop RESTful APIs and handle server-side logic.
- **PostgreSQL** – SQL database used to store user data, uploaded documents, and processed results efficiently.
- **Natural Language Processing (NLP)** – Enables automated summarization, keyword extraction, and insight generation from research papers.
- **JWT Authentication** – Provides secure user authentication and session management.
- **REST APIs** – Facilitate smooth communication between frontend and backend components.
- **Cloud Storage** – Handles secure storage and retrieval of uploaded research documents.
- **Email Integration** – Enables automated notifications and communication with users.
- **Deployment Platforms** – Used for hosting and scaling the application efficiently.



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RESEARCH PAPER ON ARTIFICIAL INTELLIGENCE & ITS APPLICATIONS

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ABSTRACT:-

It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable. While no consensus definition of Artificial Intelligence (AI) exists, AI is broadly characterized as the study of computations that allow for perception, reason and action. Today, the amount of data that is generated, by both humans and machines, far exceeds humans' ability to absorb, interpret, and make complex decisions based on that data. Artificial intelligence forms the basis for all computer learning and is the future of all computer decision making. This paper examines features of artificial intelligence, introduction, definitions of AI, history, applications, growth and achievements.

KEYWORDS: machine learning, deep learning, neural networks, Natural Language Processing and Knowledge Base System

INTRODUCTION:

Artificial Intelligence (AI) is the branch of computer science which deals with intelligence of machines where an intelligent agent is a system that takes actions which maximize its chances of success. It is the study of ideas which enable computers to do the things that make people seem intelligent. The central principles of AI include such as reasoning, knowledge, planning, learning, communication, perception and the ability to move and manipulate objects. It is the science and engineering of making intelligent machines, especially intelligent computer programs.

ARTIFICIAL INTELLIGENCE METHODS:

Machine Learning:-

It is one of the applications of AI where machines are not explicitly programmed to perform certain tasks; rather, they learn and improve from experience automatically. Deep Learning is a subset of machine learning based on artificial neural networks for predictive analysis. There are various machine learning algorithms, such as Unsupervised Learning, Supervised Learning, and Reinforcement Learning. In Unsupervised Learning, the algorithm does not use classified information to act on it without any guidance. In Supervised Learning, it deduces a function from the training data, which consists of a set of an input object and the desired output. Reinforcement learning is used by machines to take suitable actions to increase the reward to find the best possibility which should be taken in to account.

Natural Language Processing(NLP)

It is the interaction between computers and human language where the computers are programmed to process natural languages. Machine Learning is a reliable technology for Natural Language Processing to obtain meaning from human languages. In NLP, the audio of a human talk is captured by the machine. Then the audio to text conversation occurs, and then the text is processed where the data is converted into codes. Then the machine uses the audio to respond to humans. Applications of Natural Language Processing can be found in IVR (Interactive Voice Response) applications used in call centres, language translation applications like Google Translate and word processors such as Microsoft Word to check the accuracy of grammar in text. However, the nature of human language makes the Natural Language Processing difficult because of the rules which are involved in the parsing of

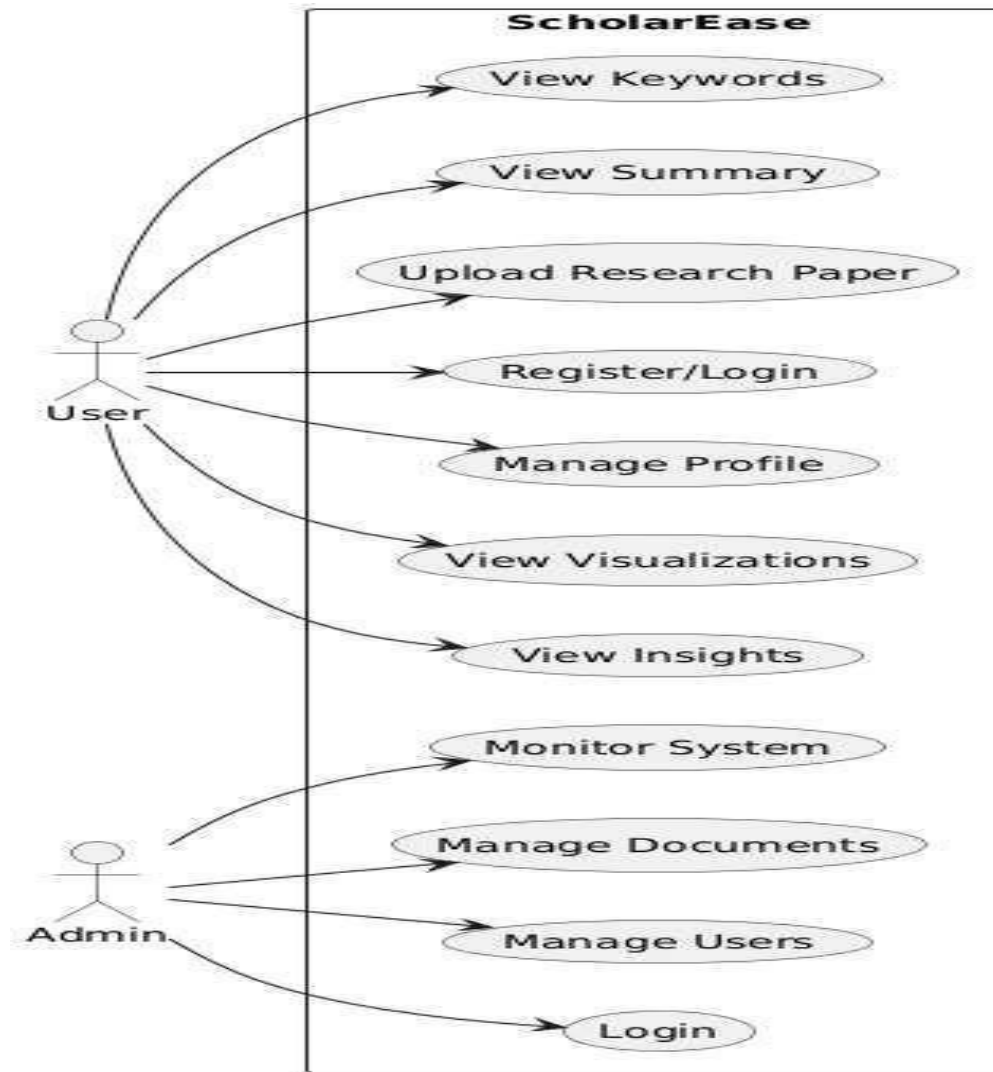
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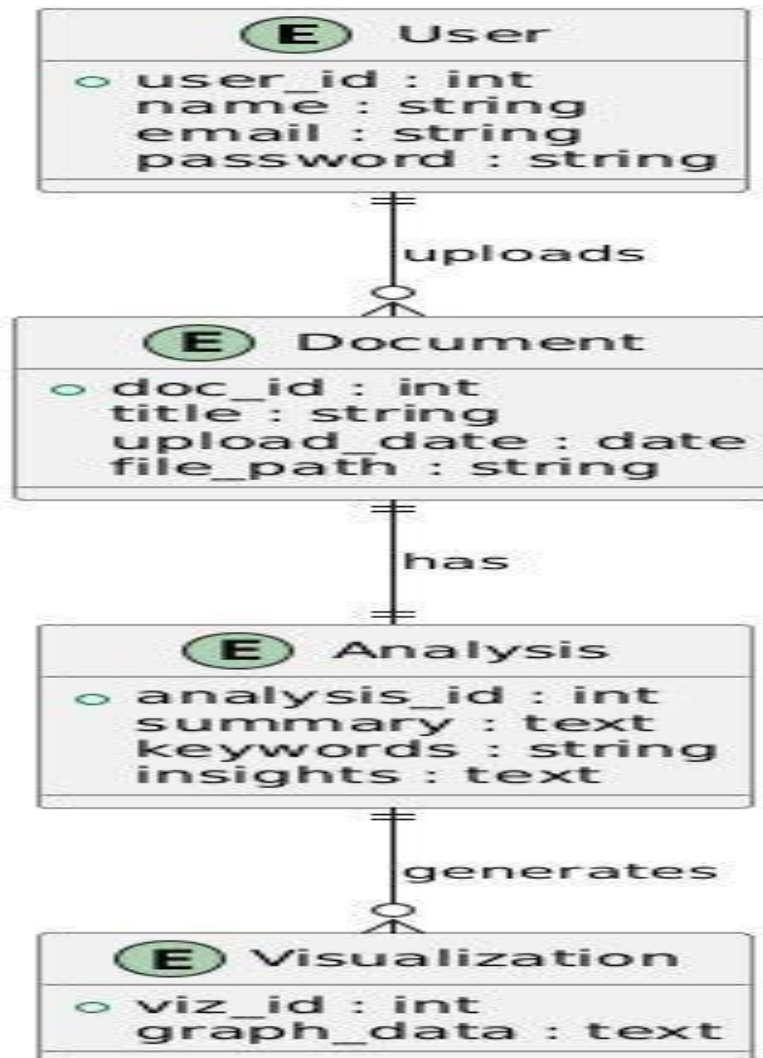
AI Simplify Comments Translate Outline

Simplifying...

Use Case Diagram



ER Diagram



The current system has certain limitations, mainly due to the use of basic NLP models, which may not always produce highly accurate summaries for complex research papers. It also supports limited document formats and lacks real-time collaboration features, while visualizations may not fully capture intricate relationships in technical content.

In the future, these limitations can be addressed by integrating advanced AI models for better contextual understanding, adding support for multiple formats, and enabling collaborative features. Enhancements such as personalized recommendations, multilingual support, and mobile application development can further improve scalability and user experience.

ScholarEase demonstrates the potential of modern web technologies and Artificial Intelligence in simplifying and enhancing academic research analysis. Developed using a robust full-stack architecture, it integrates core features such as document upload, automated summarization, keyword extraction, and insightful visualizations with advanced capabilities powered by Natural Language Processing. The platform is further strengthened by secure authentication and efficient data management, ensuring a seamless and reliable user experience. Future enhancements may include advanced AI models, collaborative features, multilingual support, and mobile application integration to improve accessibility and scalability. Overall, the project combines intelligent automation with user-centric design, offering a comprehensive solution for next-generation research assistance.